Confusion Matrices & Macro-F1

Pre-Work Module 04

ECBS5200 — Practical Deep Learning Engineering

Why accuracy lies

Imagine a dataset with 100 complaint classes. One class has 40% of all examples. The other 99 share the remaining 60%.

A model that **always predicts the majority class** gets **40% accuracy**.

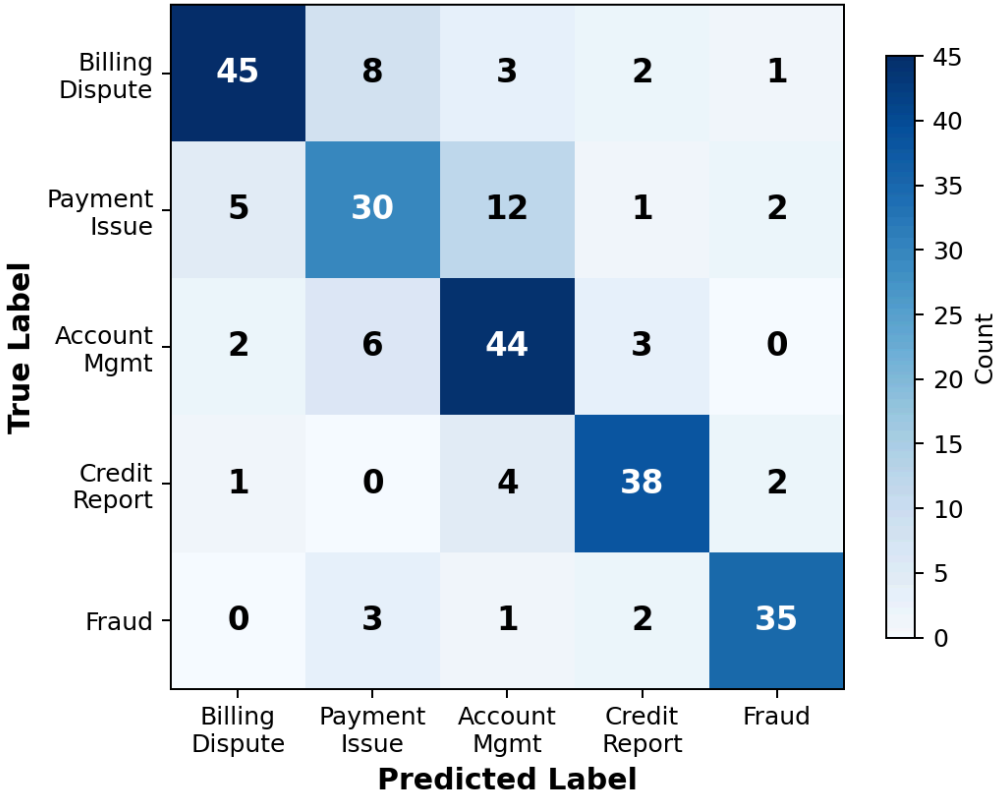
It has learned **nothing**. It can't classify a single rare complaint. But 40% accuracy sounds… not terrible?

With enough class imbalance, a "dumb" majority-class predictor can beat a model that's actually trying.

The confusion matrix

A confusion matrix shows you **exactly** what your model is doing — for every class.

- **Rows** = true labels
- **Columns** = predicted labels
- **Diagonal** = correct predictions
- Off-diagonal = mistakes



Reading a confusion matrix

	Pred: A	Pred: B	Pred: C
True: A	45	3	2
True: B	8	30	12
True: C	1	5	44

What the off-diagonal tells you:

- **Row B, Col A (8):** 8 examples of class B were misclassified as A → the model confuses B with A
- **Row B, Col C (12):** Even more B examples were called C → B is the hardest class
- **Row C, Col B (5):** Some C examples get called B → B and C look similar to the model

These patterns tell you **where to focus** — maybe B and C need better features, or more training data.

Per-class metrics: precision, recall, F1

For each class, we can compute three numbers:

Precision — "When the model predicts this class, how often is it right?"

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Recall — "Of all examples of this class, how many did the model find?"

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

F1 — The harmonic mean of precision and recall. Balanced trade-off.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Macro-F1: our primary metric

Macro-F1 = compute F1 for each class, then take the **unweighted average**.

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^{K} F1_k$$

Every class counts equally, whether it has 10,000 examples or 10.

This is our **primary evaluation metric** for the entire course.

Why? Because we want our model to work well on **all** complaint types — not just the popular ones.

Macro vs micro vs weighted

Averaging	How it works	Good when...
Macro	Average F1 across classes equally	You care about rare classes as much as common ones
Micro	Pool all TP/FP/FN, compute one F1	You care about overall correctness (≈ accuracy)
Weighted	Average F1, weighted by class size	You want a middle ground

For this course: macro.

Micro F1 is basically accuracy in disguise for multiclass problems. It lets common classes dominate. Weighted is a compromise, but it still under-weights rare classes. Macro is the fairest to all classes.

Real example: 153 raw classes, massive imbalance

Our dataset has **153 complaint categories** in the raw data.

- The **largest class** has ~7,600 examples (11.8%)
- The **smallest classes** have just 1 or 2 examples
- Some class names look suspiciously similar — we'll deal with that in Week 1

A majority-class predictor on this data:

- **Accuracy:** 11.8% (but random is only 0.65% — so 18x better than random!)
- **Macro-F1:** 0.001 (it gets 152 out of 153 classes completely wrong)

Accuracy says "not bad." Macro-F1 says "you learned almost nothing." **Macro-F1 is telling the truth.**

Key takeaways

1. **Accuracy is misleading** with imbalanced classes — it rewards ignoring rare classes
2. **Confusion matrices** show you exactly where the model is right and wrong
3. **Precision** = "are the model's predictions trustworthy?" **Recall** = "does the model find all examples?"
4. **F1** balances precision and recall; you need both to be good
5. **Macro-F1** treats every class equally — our primary metric all semester
6. In the notebook, you'll see this with your own eyes on our real dataset